

$$\begin{aligned}
 D_{1i} &= 1 && \text{if a male teacher} \\
 &= 0 && \text{otherwise (i.e., female teacher)} \\
 D_{2i} &= 1 && \text{if convent educated} \\
 &= 0 && \text{otherwise (i.e., non-convent educated)}
 \end{aligned}$$

In the above model (14.13) we have used two dummy variables one each for the two qualitative explanatory variables ‘gender of the teacher’, and ‘type of education of the teacher’. The base or the benchmark category in the above model is ‘non-convent educated female school-teacher’.

Assuming $E(u_i) = 0$, from equation (14.13), we can have the following:

Mean salary of a non-convent educated female school-teacher:

$$E(Y_i | D_1 = 0, D_2 = 0, X_i) = \alpha + \delta X_i \quad \dots (14.14)$$

Mean salary of non-convent educated male school-teacher:

$$E(Y_i | D_1 = 1, D_2 = 0, X_i) = (\alpha + \beta) + \delta X_i \quad \dots (14.15)$$

Mean salary of convent educated female school-teacher:

$$E(Y_i | D_1 = 0, D_2 = 1, X_i) = (\alpha + \gamma) + \delta X_i \quad \dots (14.16)$$

Mean salary of convent educated male school-teacher:

$$E(Y_i | D_1 = 1, D_2 = 1, X_i) = (\alpha + \beta + \gamma) + \delta X_i \quad \dots (14.17)$$

A variety of hypotheses can be tested by the Ordinary Least Squares (OLS) estimation of the model (14.13). For example, we can test for the significance of the differential intercept terms β or γ , or both β and γ thereby determining which of the possibilities exists without running the regression separately for each combination of gender and type of education.

In a similar fashion we can extend the model to include more than one quantitative and more than two qualitative variables. However, we must keep in mind that *the number of dummies for each of the qualitative variable should be one less than the number of categories of that variable.*

14.5 TESTING FOR STRUCTURAL STABILITY THROUGH DUMMY VARIABLES

In the regression models that have been discussed so far in the present unit we have considered that the qualitative variables affect the intercept term only but not the slope coefficient. But what happens if the slope coefficients are also affected by the qualitative variables? In such situations, testing the differences in the intercepts alone will be of little significance. Therefore, we need to look for a methodology that will identify whether the differences in two or more regressions are due to differences in the intercept, or slopes or both slope and intercept. In order to understand this problem, let us consider the following example. Suppose we are interested in estimating a simple savings function that relates domestic household savings (S) with the gross domestic product (Y) of India for the period

1980-81 to 2002-03. The relevant data is given in Table 10.3 below. One way of proceeding is to simply run an OLS regression of S on Y for the entire period 1980-81 to 2002-03 if the relation between savings and GDP do not change over the entire period. But it may not be so. India, in 1991 introduced a series of economic reforms thereby bringing in substantial change in its economic system. The introduction of economic reforms would have also influenced considerably the savings-income relationship. Thus, our objective is now to check whether the savings-income relationship has undergone a structural change between the two time periods or not. By structural change we mean that the parameters of the savings function have changed. Let us assume the model

$$S_t = A_1 + A_2 Y_t + u_{1t} \quad \dots (14.18)$$

for the pre-reforms period; and

$$S_t = B_1 + B_2 Y_t + u_{2t} \quad \dots (14.19)$$

for the post-reforms period

where, S_t = household savings in period t, (t = 1980-81 to 2002-03)

Y_t = gross domestic product in period t, (t = 1980-81 to 2002-03)

u = the error term

The Chow test would tell us whether there was a structural change in the saving-income relationship over the concerned period. However, it will not tell us whether the difference in the two regression models (14.18) and (14.19) is in their intercept values or in the slope values or in both. Comparing the two models (14.18) and (14.19) we see that there are four possibilities: (these possibilities are illustrated in Fig.14.3)

- $A_1=B_1$, and $A_2=B_2$; i.e., the two regressions are identical. This is the case of *coincident regression*. (refer to Fig.14.3a)
- $A_1 \neq B_1$, but $A_2=B_2$; i.e., the two regressions differ only in their location or the intercepts. This is a case of *parallel regression*. (refer to Fig.14.3b)
- $A_1=B_1$, but $A_2 \neq B_2$; i.e., the two regressions have same intercept term but different slopes. This is a case of *concurrent regression*. (refer to Fig.14.3c) and
- $A_1 \neq B_1$, and $A_2 \neq B_2$; i.e., the two regressions are completely different. This is a case of *dissimilar regression*. (refer to Fig.14.3d)

From the data on household savings and gross domestic product for India given in Table 14.2 we can run the regressions (14.18) and (14.19) and apply the Chow test to see whether the savings function has undergone a structural change between the two time periods. (Students are advised to apply the Chow Test and find out the result as an assignment)

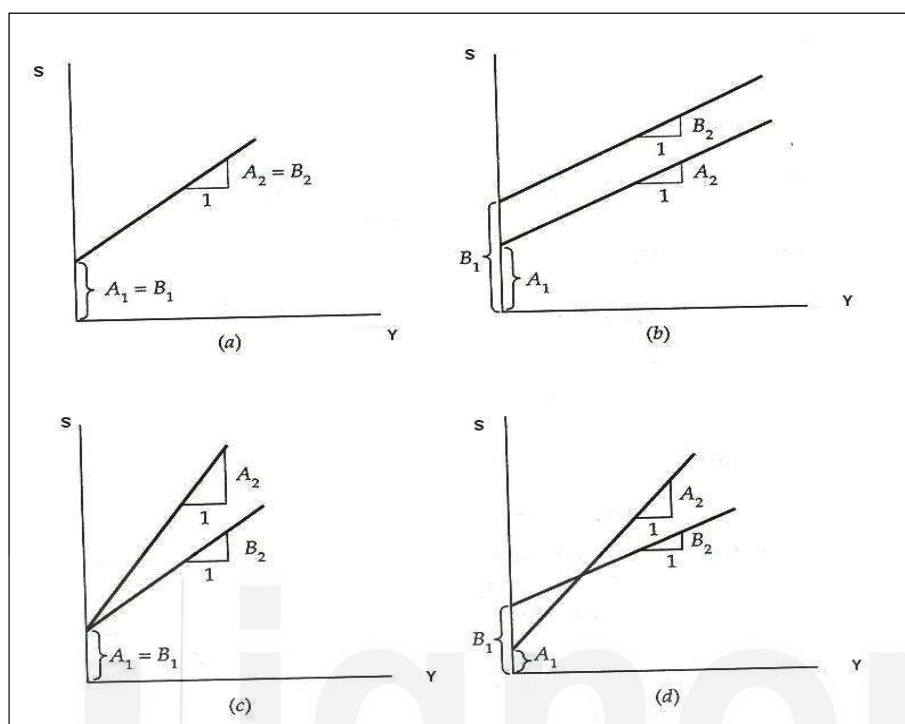


Fig. 14.3: Possible Differences in the Two Regression Models

Table 14.3: Domestic Savings and GDP in India, 1980-81 to 2002-03

(at constant prices) **(Rs. crore)**

Year	Domestic Savings (S)	Gross Domestic Product (Y)	Year	Domestic Savings (S)	Gross Domestic Product (Y)
1980-81	27136.0	401128.0	1992-93	162906.0	737791.6
1981-82	31355.0	425072.8	1993-94	193621.3	781345.0
1982-83	34368.0	438079.5	1994-95	251463.4	838031.0
1983-84	38587.0	471742.2	1995-96	298747.3	899563.0
1984-85	46063.0	492077.3	1996-97	317260.7	970082.0
1985-86	54167.0	513990.0	1997-98	352178.0	1016595.0
1986-87	58951.0	536256.7	1998-99	374659.0	1082748.0
1987-88	72908.0	556777.8	1999-00	468681.0	1148368.0
1988-89	87913.0	615098.4	2000-01	495986.0	1198592.0
1989-90	106979.0	656331.2	2001-02	535185.0	1267833.0
1990-91	131340.0	692871.5	2002-03	597697.0	1318321.0
1991-92	143908.0	701863.2			

An alternative way of testing whether the savings function has undergone a structural change or not is the use of *dummy variables*. Let us see how the dummy variables can be used to handle the problem of structural change.

Let us write the savings function as

$$S_t = a + bD_t + cY_t + d(D_tY_t) + u_t \quad \dots (14.20)$$

where, S_t = household savings in period t

Y_t = GDP in period t

$D_t = 1$ for observation in the pre-reform period

= 0 otherwise, i.e., for observation in the post-reform period

In model (14.20) b is the *differential intercept* and d is the *differential slope coefficient* indicating how much the slope coefficient of the pre-reform periods savings function differs from the slope coefficient of the savings function in the post reform period. Just as the introduction of an additive dummy enables us to distinguish between the intercepts of the two periods, the introduction of multiplicative dummy enables us to differentiate between the slope coefficients in the two periods.

Let us see the implications of the regression model (14.20). Assuming $E(u_t) = 0$, we obtain

$$E(S_t|D_t = 0, Y_t) = \alpha + cY_t \quad \dots (14.21)$$

$$E(S_t|D_t = 1, Y_t) = (\alpha + b) + (c + d)Y_t \quad \dots (14.22)$$

The above two equations are the mean savings function for the pre-reforms period and the post-reforms period respectively as represented by model (14.18) and (14.19) with $a=A_1$, $c=A_2$, and $(a+b)=B_1, (c+d)=B_2$. Thus, with the help of dummy variables a single regression can easily be used to obtain two sub-period regressions.

Now estimating the regression (14.20) using the savings and income data given in Table 14.3 we obtain

$$\hat{S}_t = -133690.9 - 228328.9 D_t + 0.375 Y_t + 0.3385 D_t Y_t \quad \dots (14.23)$$

$$SE \quad (21259.0) \quad (30775.09) \quad (0.0386) \quad (0.0441)$$

$$t\text{-stat} \quad (-6.289) \quad (-7.429) \quad (9.717) \quad (7.674)$$

$$R^2 = 0.9954$$

We can see from the estimated regression model (14.23) that both the differential intercept and the differential slope coefficients are statistically significant. It implies that the regression models or the savings-income relationship for both the periods are different. Both the regression models can be diagrammatically represented by Fig.14.3d. Substituting for D_t the values 0 and 1 we obtain two separate equations as given below.

For the pre-reforms period, when $D_t = 0$, we have

$$\hat{S}_t = -133690.9 - 228328.9 (0) + 0.375 Y_t + 0.3385(0) Y_t$$

$$\text{Or, } \hat{S}_t = -133690.9 + 0.375 Y_t \quad \dots (14.24)$$

For the post-reforms period we have

$$\hat{S}_t = -133690.9 - 228328.9(1) + 0.375 Y_t + 0.3385(1) Y_t$$

$$\begin{aligned} \text{Or, } \hat{S}_t &= -133690.9 - 228328.9(1) + 0.375 Y_t + 0.3385(1) Y_t \\ &= -362319.8 + 0.7135 Y_t \quad \dots (14.25) \end{aligned}$$

(Note: You can see that these regression models are the same as those obtained from the Chow test procedure).

Thus, from the above discussion we can infer that the dummy variable technique has certain advantages over the Chow test.

- The dummy variable technique requires running only a single equation while for the Chow test one has to run a number of regressions, one each for each of the periods concerned.
- The Chow test does not tell us whether the intercept or the slope coefficient or both are different in the two sub-samples. In fact, we cannot tell by the Chow test which of the four possibilities presented in Fig.14.3 exists. In this respect the use of dummy variables approach has a definite advantage. It not only tells whether the two regressions are different but also pinpoints the source of their difference – whether it is because of the intercept or the slope or both the coefficients.
- The use of a single regression equation as in the case of dummy variable approach would mean higher degrees of freedom compared to more than one regression model (in the case of Chow test). Thus, the dummy variable technique improves the relative precision of the estimated parameters.

14.6 USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

Many economic time series based on monthly or quarterly data exhibit seasonal patterns. For example, the sale of woollens during the winter months, sales of departmental stores during the festive season, sale of soft drinks in the summer months, sale of refrigerators and air-conditioners in the summer season etc are some of the examples of time series that show seasonal pattern. It is often desirable to eliminate the seasonal component of such times-series. The process of removing/eliminating the seasonal factor is known as de-seasonalisation or seasonal adjustment. The adjusted time series thus obtained is called the seasonally-adjusted, or the de-seasonalised times series.

There are many ways of deseasonalising a time series. Dummy variables can also be effectively used for deseasonalising a time series. Let us deal with the use of dummy variable method in deseasonalisation of time series. Let us consider the sales of a departmental store. Table 14.4 gives data on the quarterly sales and profits of a departmental store in New Delhi during the period 2021-22 and 2024-25.

Table 14.4: Sales and Profits of a Departmental Store (Rs. Lakhs)

Year	Quarter	Profit (Y)	Sales (X)
2021-22	I	16.63	132.1
	II	19.41	136.5
	III	24.01	177.7
	IV	19.91	157.5
2022-23	I	17.55	150.7
	II	20.90	152.4
	III	25.61	199.8
	IV	21.97	173.0
2023-24	I	19.46	167.4
	II	22.14	179.1
	III	27.42	227.0
	IV	22.72	191.6
2024-25	I	21.42	185.0
	II	25.41	187.3
	III	32.07	261.5
	IV	25.49	219.2

The profit function of the departmental store is dependent on the sales. Let us represent the profit function as

$$Y_t = A_1 + A_2D_{2t} + A_3D_{3t} + A_4D_{4t} + A_5X_t + u_t \quad \dots (14.26)$$

where, Y_t = profit of the store in period t

X_t = sales in period t

The seasonal dummies are defined as

$$D_{2t} = 1 \text{ if the profit lies in the second quarter}$$

= 0 otherwise

D_{3t} = 1 if the profit lies in the third quarter
= 0 otherwise

D_{4t} = 1 if the profit lies in the fourth quarter
= 0 otherwise

In the above model we have, through the four quarters, incorporated seasonal variation. It is assumed that the qualitative variable 'season' has four categories. Therefore, it requires the use of three dummy variables (D_{2t}, D_{3t}, D_{4t}) in the regression model. Note that the first quarter has been taken as the base or benchmark quarter. We have used intercept dummy variables only. Each of these differential intercepts tells us the extent by which the mean value of Y (i.e., the mean profit) differs in each quarter compared to the base quarter. Using the data in Table 14.4, the regression results of equation (14.26) are as follows:

$$\hat{Y}_t = 3.9573 + 2.7314D_2 + 3.1321D_3 + 1.2841D_4 + 0.0932X_t \quad \dots (14.27)$$

SE	(1.195)	(0.491)	(0.642)	(0.525)	(0.007)
t-value	(3.313)	(5.568)	(4.880)	(2.445)	(12.95)

$R^2 = 0.977$

We observe from the t-values that all the differential intercepts are statistically significant. It implies that the average profits differ across each of the four quarters. It shows that the average profit is the maximum in the third quarter (October-December) when the sales are the highest on account of the festive season in the country.

In the regression model (14.27) assumed that the seasonal variations affect the intercept only and not the slope parameter. But it may not be so in reality. To find out whether the seasonal variation have effects on the intercept or the slope (or both), we use the technique of differential intercept and differential slope coefficient discussed in the previous section (model 14.16). Applying the method we can rewrite model (14.26) as shown by regression model (14.28) and test for the significance of differential intercept and differential slope terms.

$$Y_t = A_1 + A_2D_{2t} + A_3D_{3t} + A_4D_{4t} + A_5X_t + A_6(D_{2t}X_t) + A_7(D_{3t}X_t) + A_8(D_{4t}X_t) + u_t \quad \dots (14.28)$$

where, the differential slope coefficients A_6 , A_7 and A_8 tell us the amount by which the slope coefficient of the second quarter, the third quarter and the fourth quarter differ from the base or the first quarter respectively.

14.7 POOLING CROSS SECTION AND TIME SERIES DATA

Let us consider Table 14.5 which gives data on the energy demand (in million tons of oil equivalent) and value of output (in Rs. million) for three sectors of the Indian economy. This is an example of a cross section time series data where we are interested in finding out the relation between energy demand and value of output for three sectors of the economy and for each sector we have data for 18 years from 2005-06 to 2023-24.

There are several ways to study the relationship between energy demand and value of output. First, we can run the following regression models for each sector separately:

Let us consider

$$Y_t = A_1 + A_2 X_t + u_t \quad \dots (14.29)$$

for the agricultural sector;

$$Y_t = B_1 + B_2 X_t + u_t \quad \dots (14.30)$$

for the industrial sector; and so on.

where, Y_t = energy consumption

X_t = sectoral value of output

Using the dummy variable technique as discussed earlier or the Chow test we can find out if the parameters of these demand functions are the same or not.

The second way is to estimate for each year, the cross-sectional regression. In such a case there would be one regression for each of the 18 years giving a total of 18 regressions to be estimated.

The third way is to pool all the 54 observations (18 times series observations for the three sectors) and estimate the following regression

$$Y_{it} = A_1 + A_2 D_{1t} + A_3 D_{2t} + A_4 X_{it} + u_{it} \quad \dots (14.31)$$

where, i stands for i -th sector and t for the t -th time period.

In (14.31) we assumed that only the intercept terms differ across the sector but not the slope terms. You can assume both the slope coefficients and intercept terms to be different across sectors and test of their significance themselves).

Estimation of (14.31) using data in Table 14.5 gives us

$$\hat{Y}_t = 13.133 - 17.426 D_{1t} + 6.696 D_{2t} + 0.0004 Y_t \quad \dots (14.32)$$

Std error (1.482) (1.682) (3.010) (0.0000)

t -statistic (9.195) (-10.356) (2.224) (18.195)

$R^2 = 0.97602$

The above results show that both the dummies are statistically significant. It implies that the three sectors have different intercept terms (As an exercise you

should incorporate slope dummies and test whether they are also significant or not).

Table 14.5: Energy Demand and Value of Output

Year	<i>Agricultural Sector</i>		<i>Industrial Sector</i>		<i>Transport Sector</i>	
	Energy demand	Value of output	Energy demand	Value of output	Energy demand	Value of output
	(mtoe)	(Rs. Mn)	(mtoe)	(Rs. Mn)	(mtoe)	(Rs. Mn)
05-06	5.47	466490	49.63	950820	18.04	249630
06-07	5.67	491390	54.74	1026470	18.45	263940
07-08	6.60	483580	58.32	1064180	19.46	271850
08-09	6.80	535250	61.64	1150020	20.39	283070
09-10	7.80	535430	63.63	1218210	21.09	302310
10-11	8.56	547970	67.71	1274720	22.53	324270
11-12	10.46	533600	72.80	1362240	23.67	344410
12-13	13.86	535290	72.24	1452530	25.55	368640
13-14	15.04	622140	79.13	1586490	27.23	384790
14-15	16.38	632630	83.02	1750530	28.83	410240
15-16	18.81	656530	86.42	1886010	29.09	428940
16-17	21.78	641180	90.34	1875600	30.72	455950
17-18	23.72	680090	93.76	1949940	31.83	477090
18-19	26.62	705130	97.43	2051620	33.35	511310
20-21	29.40	722640	102.17	2241360	35.01	560170
21-22	32.70	703150	101.10	2523590	38.67	623170
22-23	32.82	758880	109.51	2702180	41.65	674410
23-24	34.29	823410	112.24	2817880	43.45	727850

- 1) Consider the data given in Table 14.3 and divide it into two sub-periods 1980-81 to 1991-92 and 1992-93 to 2002-03. Test for structural break in the savings functions using Chow test.

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- 2) Use the data given at Table 14.3 and run the regression given at equation (10.20). Are the differential intercept and differential slope coefficients statistically significant?

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14.8 LET US SUM UP

In many instances, in regression equations, we have explanatory variables which are qualitative in nature. It is difficult to quantify these qualitative variables as they at best can be divided into certain categories. In such cases we use the dummy variable model.

The dummy variable can affect the intercept or slope or both. Accordingly, we take intercept or slope dummies. Remember that dummy variables are used, as are explanatory variables, on the basis of the logic we build up. Thus, behind every regression model there is a theoretical basis.

Dummy variables can be used in seasonal analysis. It also can be used in pooling cross-sectional and time series data.

14.9 KEY WORDS

ANOVA Models: This is similar to the case where only dummy variable is taken as explanatory variable.

ANCOVA Models: This is similar to regression model having both qualitative and quantitative variables as explanatory variables.

Cross-section Data: This refers to data measured over geographically dispersed units at a point of time.